$t=0^-$: inductor : Shorted, capacitor: open $\Rightarrow v_C=-20$, $v_L=0$, $v_R=0$, $i_L=0$

$t=0^+$: inductor current can't change fast $\Rightarrow i_L(0^+)=0$

voltage across capacitor can't change fast $\Rightarrow v_C(0^+)=-20V$, $v_R(0^+)=4V$

$t=\infty$: steady state (inductor shorted, capacitor open) $\Rightarrow i_L(\infty)=1A$, $v_R(\infty)=4V$, $v_C(\infty)=-20V$

for $i_L'(0^+)$: $0.6\frac{di_L}{dt}=v_C \Rightarrow i_L'(0^+)=0$

EE2015: Electric Circuits and Networks

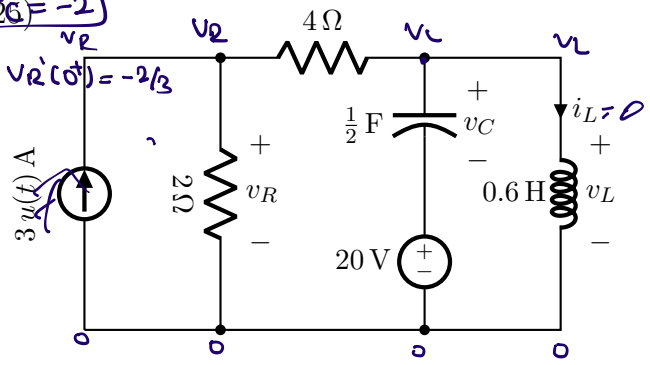
Tutorial 6

(September 26, 2026)

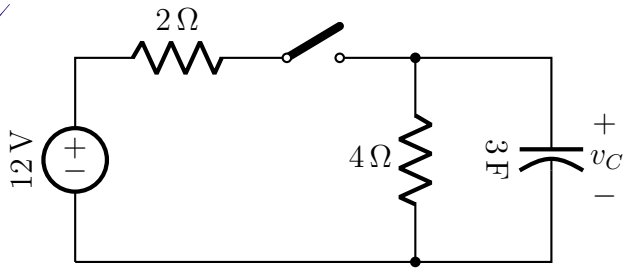
$i_L+\frac{1}{2}\frac{dv_C}{dt}=\frac{v_L-v_R}{4} \Rightarrow 0+\frac{1}{2}v_C'(0^+)=\frac{0-4}{4}=-1 \Rightarrow v_C'(0^+)=-2$

$3=\frac{v_R}{2}+\frac{v_R-v_C-20}{4} \Rightarrow 0=\frac{v_R'(0^+)}{4}-\frac{v_C'(0^+)}{4} \Rightarrow v_R'(0^+)=-2/3$

1. For the circuit shown on the right find: (a) $i_L(0^+)$, $v_C(0^+)$, $v_R(0^+)$, (b) $i_L'(0^+)$, $v_C'(0^+)$, $v_R'(0^+)$, and (c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$.



2. For the circuit shown on the left find $v_C(t)$ if



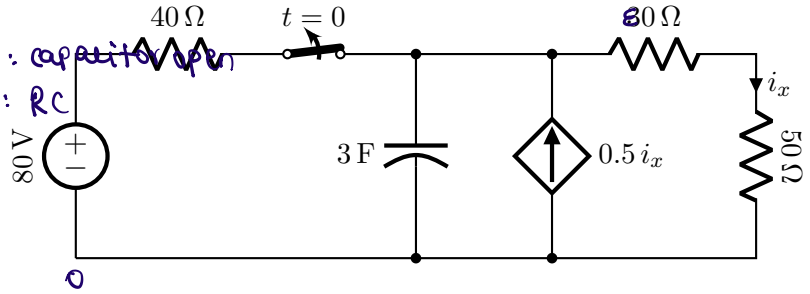
(a) The switch has been open for a long time and is closed at $t = 0$.

(b) The switch has been closed for a long time and is opened at $t = 0$.

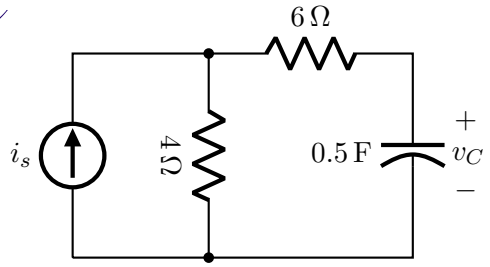
3. For the circuit shown on the right find $i_x(t)$ for all t.

$t<0$: capacitor open

$t>0$: RC



4. For the circuit shown on the left it is given that $v_C(0^-) = 0$. A current pulse $i_s(t) = 2 \cdot [u(t) - u(t-1)]$ A is applied to the circuit. Find $v_C(t)$ for $t \geq 0$.

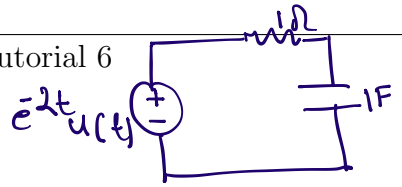


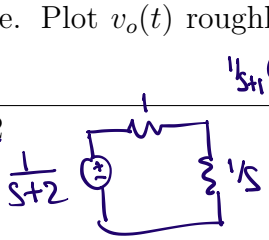
10

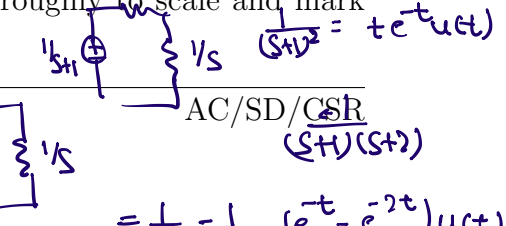
5. (a) A voltage source $v_s(t) = e^{-2t}\,u(t)$ is applied across a series combination of $1\,\Omega$ and $1\,F$. Find $v_c(t)$ for $t \geq 0$.

(b) Repeat for $v_s(t) = e^{-t}\,u(t)$. Compare this answer with the previous one and comment on the result.

6. Consider the two circuits shown below. Derive the expressions for $v_o(t)$ when $v_i(t) = V_p\,u(t)$. The capacitors have zero initial charge. Plot $v_o(t)$ roughly to scale and mark

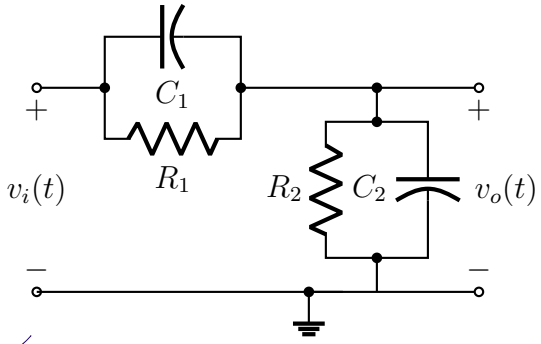


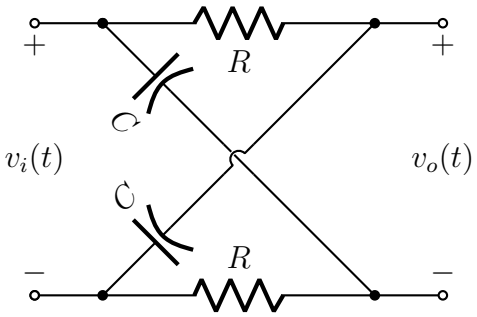




$V_i(s) =$

the important points, including the time-constants. Next, open-circuit the capacitors and calculate $v_o(t)$. How are these values related to $v_o(t)$ calculated previously?





7. In the circuit shown below, the rotary switch S rotates at 10 rpm. It makes contact alternately with the conducting segments A and B for 3 s each. The transition from A to B happens intantaneously. Assume that S has been rotating for a long time.

   (a) Plot $i_L(t)$, clearly marking the maximum and minimum values.

   (b) Calculate the maximum and minimum voltage across the 2 Ω resistance, and the total energy consumed by this resistance during each revolution of S.

$I_2 = I_1 + (3 - I_1)(1 - e^{-t/3})$

$t = 3 \Rightarrow I_2 = I_1 + (2 - I_1)(1 - 1/e)$

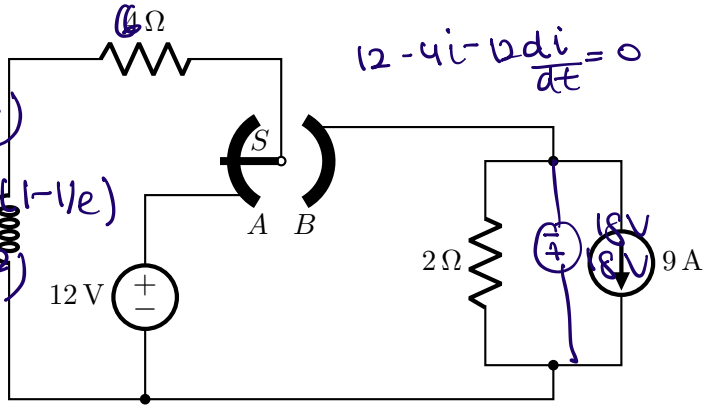
$I_1 = I_2 + (3 - I_2)(1 - e^{-t/2})$

$12 - 4i - 12\frac{di}{dt} = 0$

$3(1 - e^{-t/3})$

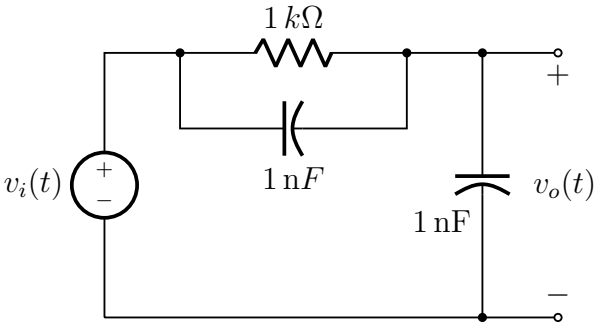
$3(1 - 1/e)$ +

$3(1 - e^{(t-1)/2})$



8. The circuit shown on the right is excited by $v_i(t)$, where $v_i(t) = \sum_{k=0}^{\infty} x(t - kT_0)$. The waveform $x(t)$ is defined as follows:

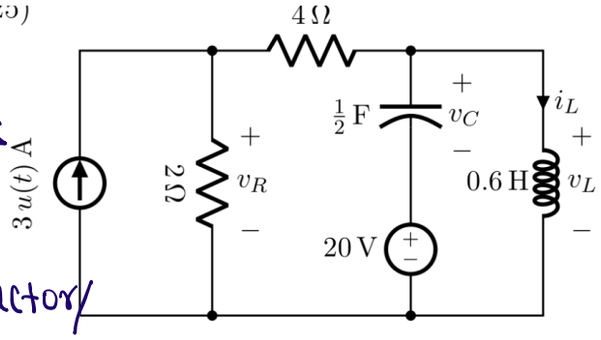
$$x(t) = \begin{cases} 1 & 0 < t < T_0/2 \\ -1 & T_0/2 < t < T_0 \\ 0 & \text{otherwise} \end{cases}$$



where $T_0 = 2\,\mu$s. Sketch $v_o(t)$ after steady-state has been reached and mark the important points.

1. 0A −20V +4V

For the circuit shown on the right find: (a) $i_L(0^+)$, $v_C(0^+)$, $v_R(0^+)$, (b) $i'_L(0^+)$, $v'_C(0^+)$, $v'_R(0^+)$, and (c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$. → 0A/s 2V/s → 4 2V/s 1A −20V



(a) $t<0 \Rightarrow i=0 \Rightarrow$ no current in inductor / capacitor

(c) $t=\infty \Rightarrow$ capacitor = open circuit $\Rightarrow v_C(\infty) = -20V$

inductor = shorted

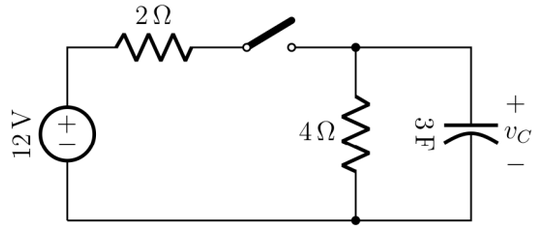
$R_{eff} = 8/6 = 4/3 \rightarrow v_L(\infty) = 3 \cdot 4/3 = 4V = v_R(\infty)$

$i_L(\infty) = 4/4 = 1A$

(b) $i_C + i_L = \frac{v_R - v_L}{4}$; $i_C = C\frac{dv_C}{dt} = C\,dv_R/dt$ $\quad v_L(t) = L\,di(t)/dt$ $\quad v_L(0^+) = v_R(0^+) = 0 \Rightarrow i_L(t) = 0$

at 0^+, $v_R - v_L/4 = -1A$; $i_L(0^+) = 0 \Rightarrow i_C(0^+) = 1A \Rightarrow i_C(0^+) = v_R(0^+) = 1/1/2 = 2V/s$

2.

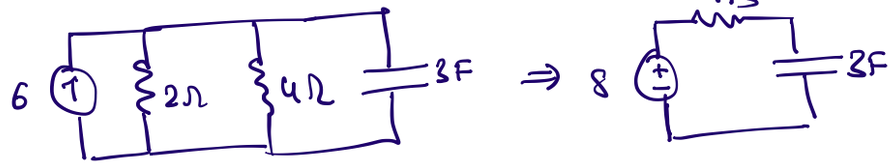


For the circuit shown on the left find $v_C(t)$ if

(a) The switch has been open for a long time and is closed at $t = 0$.

(b) The switch has been closed for a long time and is opened at $t = 0$.

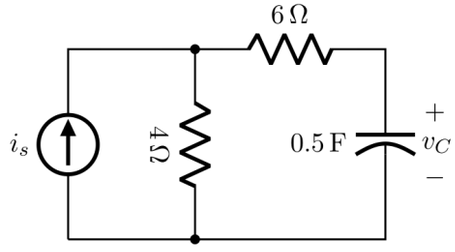
(a)



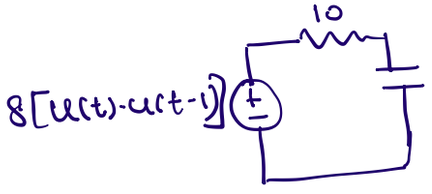
$v_C(t) = 8(1 - e^{-t/4})$

(b) $V_0 = 8V \Rightarrow V = 8e^{-t/12}$

4.



For the circuit shown on the left it is given that $v_C(0^-) = 0$. A current pulse $i_s(t) = 2 \cdot [u(t) - u(t-1)]$ A is applied to the circuit. Find $v_C(t)$ for $t \geq 0$.

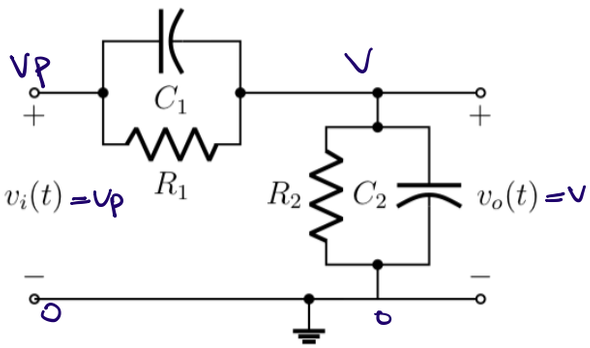


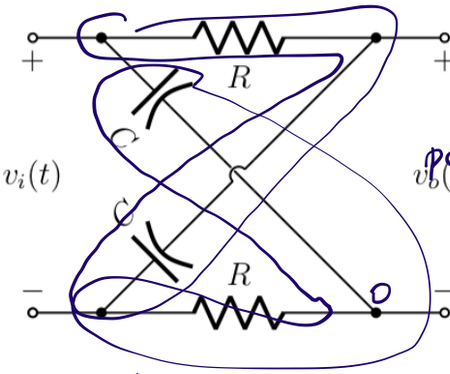
$\Rightarrow$ $t \in [0, 1) \Rightarrow v_C(t) = 8(1 - e^{-t/5})$

$t > 1 \Rightarrow v_C(t) = 8(1 - e^{-1/5})\, e^{-(t-1)/5}$

6. Consider the two circuits shown below. Derive the expressions for $v_o(t)$ when $v_i(t) = V_p\, u(t)$. The capacitors have zero initial charge. Plot $v_o(t)$ roughly to scale and mark

the important points, including the time-constants. Next, open-circuit the capacitors and calculate $v_o(t)$. How are these values related to $v_o(t)$ calculated previously?





$$\frac{Vp - V_o}{R_1} + C_1 \frac{d(Vp - V_o)}{dt} = \frac{V_o}{R_2} + C_2 \frac{dV_o}{dt}$$

$$\Rightarrow \frac{V_i}{R_1} + C_1 \frac{dV_i}{dt} = (C_1 + C_2)\frac{dV_o}{dt} + V_o\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$\frac{V_i(s)}{R_1} + sC_1 V_i(s) = s(C_1 + C_2) V_o(s) + V_o(s)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$\Rightarrow \frac{V_o(s)}{V_i(s)} = \frac{1/R_1 + sC_1}{s(C_1 + C_2) + (1/R_1 + 1/R_2)}$$

$$V_o(s) = \frac{Vp/sR_1 + C_1}{s(C_1 + C_2) + (1/R_1 + 1/R_2)}$$

$$V_i = (R + 1/sC)\, I/2 \Rightarrow I = \frac{2V_i(sC)}{sCR + 1}$$

$$V_o = V_i - IR = Vp/s - \frac{2Vp}{sRC + 1/RC}$$

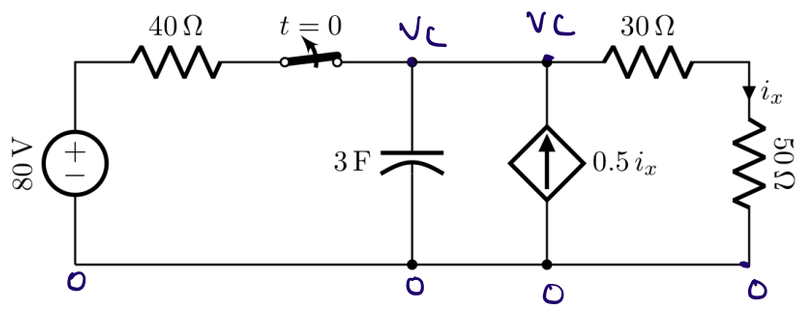
$$Vp\,u(t) - 2Vp e^{-t/RC} u(t)$$

R_1 $1+C_2$ L

$$\ln\left|\frac{Vp}{R_1(C_1 + C_2)} - \frac{v}{\tau}\right| = -t/\tau$$

$\Rightarrow$

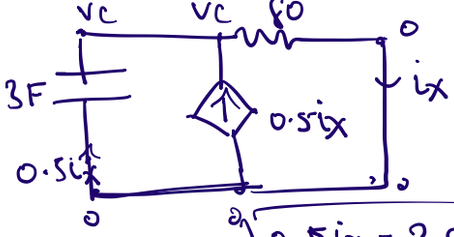
3. For the circuit shown on the right find $i_x(t)$ for all t.



$t<0 \Rightarrow$ capacitor open

$$t<0: \frac{80 - V_c}{40} - i_x + 0.5 i_x = 0 \qquad i_x = V_c/80 \Rightarrow i_x = 0.8 \qquad V_c = 64$$

$t>0$:



$0.5 i_x = 3 \cdot \frac{dv_c}{dt}$

$i_x = v_c/80$

$0.5 \cdot \frac{v_c}{80} = -3 \cdot \frac{dv_c}{dt} \Rightarrow v_c = v_0 e^{-t/480}$

$v_0 = 64$

$i_x = v_c/80 = 0.8\, e^{-t/480}$